Concurrent Dispatch for Home Energy Systems

Daniel Murong — Intro to Distributed Systems, Spring 2026

1. Introduction

A home with rooftop solar panels, a home battery, and a connection to the electric grid must have a process of routing energy among them and the household load. Spontaneous load spikes can occur, and energy prices from the grid vary throughout the day. This project implements a Go simulator of such a home, where each subsystem is a **goroutine** communicating over **channels**, and uses it to compare three battery-management policies under Duke Energy's NC Residential Solar TOU rates.

Thesis: Greedy single-variable policies are unstable across seasons. Each naive baseline wins in one season and loses in the other. A two-parameter "structural slack" policy beats both in every regime, without any forecasting. The same principle is why production auto-scalers like the Kubernetes Horizontal Pod Autoscaler and AWS Auto Scaling Groups use minimum and maximum bounds rather than a single target.

2. System Architecture

Each physical subsystem is an independent goroutine exchanging typed messages over buffered channels: `Clock --Tick--> {Solar, Load} --> Coordinator --Decision--> Battery --> Logger`. A **Clock** goroutine emits a `Tick` every simulated 5 minutes (288 a day), carrying the logical timestamp so no component reads a shared wall clock. **Solar** and **Load** goroutines consume ticks from fanned-out channels and produce independent kW values. The **coordinator** invokes the active **policy** function to produce a `Decision` (how energy flows from each source to each sink), applies it to the mutable **battery** state, and forwards the result to a **logger**. This mirrors a real distributed control plane: sensors, actuators, and a controller as independent processes over a message bus, not a single tight loop.

Inputs are calibrated against Duke Energy Schedule RSTC, NREL TMY3, the Tesla Powerwall 2 datasheet, and EIA RECS + NREL ResStock.

3. Policies Under Test

Policy	Rule
Greedy Cheap-Fill	Grid-charge to 100% during off-peak hours.
Greedy Discharge	Drain to 0% during on-peak hours.
Two-Threshold	Maintain reserve floor (20%) and charge ceiling (80%).

4. Results

Each policy was run for one simulated day under both summer and non-summer tariffs. Duke's on-peak window differs by season: **summer is 3–6 PM** (coinciding with solar's tail) and **non-summer is 6–9 AM** (*preceding* solar). This single difference drives the entire result.

Season	Policy	Net Cost	Solar Util.	Min SOC	Spilled
Summer	**Two-Threshold**	**$1.62**	63.8%	53.1%	15.9 kWh
Summer	Greedy Cheap-Fill	$2.26	57.6%	53.1%	18.6 kWh
Summer	Greedy Discharge	$2.63	36.6%	16.6%	27.9 kWh
Non-summer	**Two-Threshold**	**$2.03**	50.0%	30.6%	15.2 kWh
Non-summer	Greedy Discharge	$2.30	37.0%	33.7%	19.1 kWh
Non-summer	Greedy Cheap-Fill	$3.17	35.3%	53.1%	19.7 kWh

The greedy ranking flips between seasons. In summer, cheap-fill beats discharge by \$0.37; in non-summer, discharge beats cheap-fill by \$0.87. Two-threshold wins both, by 28% and 36% over the next-best greedy.

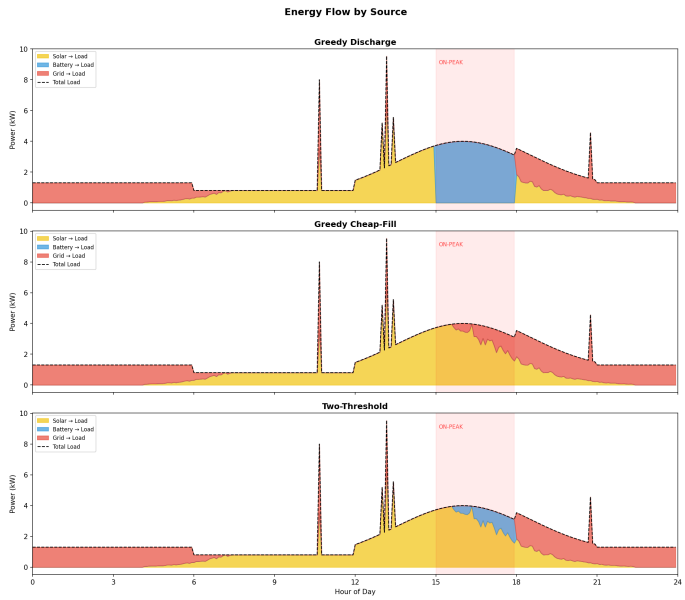

Figure 1: Summer energy flows by policy. Yellow = solar→load, blue = battery→load, red = grid→load; shaded = 3–6 PM on-peak window.

Why the flip: In summer, on-peak coincides with solar's tail (Fig. 1): cheap-fill is full at noon and dumps midday solar at \$0.03/kWh, while discharge drains to 17% SOC by mid-peak and refuses to store solar it plans to export. In non-summer, the 6–9 AM peak *precedes* solar, so draining is rational (no storage opportunity cost) and cheap-fill is catastrophic since it never discharges aggressively, so the morning peak is served from the grid at \$0.224/kWh. Two-threshold is the only policy that both covers the peak from the battery *and* leaves overnight headroom for midday solar, in either season.

5. Discussion: Structural Slack as a Distributed-Systems Primitive

Two-threshold uses two parameters, each preventing a failure point: a **reserve floor (20%)** that guarantees backup capacity, and a **charge ceiling (80%)** that preserves headroom for incoming solar. Critically, **neither threshold requires prediction**. The policy does not model when solar will arrive, how large the peak will be, or what season it is. It simply refuses to commit the last 20% of capacity in either direction, and that single structural constraint beats both greedy baselines in both seasons.

Production auto-scalers use the same principle. *Greedy cheap-fill* is **over-provisioning**: pre-scaled capacity sitting idle, organic spikes served at on-demand premium. *Greedy discharge* is **aggressive scale-down**: min-replicas set too low, producing cold starts, pool exhaustion, and cascading timeouts on the next spike. *Two-threshold* is comparable to the **K8s HPA utilization band** and **AWS ASG `MinSize`/`MaxSize`**: the lower bound offers insurance for sudden spikes in work, the upper bound limits blast radius and wasteful spending on excess capacity, neither requires forecasting.

6. Limitations & Reproduction

Single-day, single-seed, parametric solar; round-trip efficiency, the \$14/mo customer charge, and weekend tariffs are not modeled. AI assistance (Anthropic Claude) was used for code scaffolding and editorial revision; modeling and analysis are my own.